

Dr. Ahmed Reda Abdelmoneim Mohamed

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🎓 [Google Scholar](#) [LinkedIn](#)

Nationality: Egyptian

Professional Profile

Ahmed Reda Mohamed (Member, IEEE) received the Ph.D. degree in Electronic Science and Technology from Shanghai Jiao Tong University (QS: 36), China, in 2021, where he conducted research at the MoE Key Laboratory of Artificial Intelligence. He is currently a Postdoctoral Fellow with the Interdisciplinary Research Center for Communication Systems and Sensing (IRC-CSS), King Fahd University of Petroleum and Minerals (QS: 63), Saudi Arabia. His research interests include low-power analog/mixed-signal CMOS IC design, biomedical sensor interfaces, connected sensing platforms, energy harvesting, and analog IC design automation. He received the **IEEE ISSCC 2022 Student Research Preview Poster Award** and the **KFUPM Postdoctoral Excellence Award (Runner-up with Distinction, 2025–2026)**. He has authored or co-authored 29 journal and conference publications, contributed to seven patent disclosures, and served as a Research Team Member and Co-PI on competitively funded research projects supported by IRC-CSS and the KFUPM Deanship of Research.

Areas of Interest

- Low-power analog/mixed-signal CMOS ICs and biomedical sensor interfaces
- Sensors, data acquisition, and system integration
- Firmware and hardware/software co-design
- IoT, wearable, and connected sensing platforms
- Analog IC design automation, EDA, and PVT-robust optimization
- Energy harvesting, power management, and AI hardware

Work Experience

Postdoctoral Fellow (Full-Time) <i>King Fahd University of Petroleum & Minerals (KFUPM)</i>	Sept 2024 – Present <i>Dhahran, Saudi Arabia</i>
Assistant Professor (Part-Time) <i>University of Hertfordshire (UH)</i>	Sept 2023 – Sept 2024 <i>New Capital, Cairo, Egypt</i>
IC Designer and CAD Tool Developer <i>Efabless</i>	June 2022 – March 2024 <i>USA (Remote)</i>
Assistant Professor (Part-Time) <i>Egyptian Chinese University (ECU)</i>	Feb 2022 – Sept 2024 <i>Cairo, Egypt</i>
Assistant Professor (Full-Time) <i>Zagazig University (ZU)</i>	Sept 2021 – Sept 2024 <i>Zagazig, Egypt</i>
Ph.D. Researcher (Full-Time) <i>Shanghai Jiao Tong University (SJTU)</i>	Sept 2016 – Sept 2021 <i>Shanghai, China</i>
Assistant Lecturer / Researcher (Full-Time) <i>Zagazig University (ZU)</i>	Mar 2014 – Sept 2016 <i>Zagazig, Egypt</i>
Demonstrator (Full-Time) <i>Zagazig University (ZU)</i>	Nov 2009 – Mar 2014 <i>Zagazig, Egypt</i>
Design and Layout Engineer (Part-Time) <i>Technology Innovation and Entrepreneurship Centre (TIEC)</i>	Jan 2012 – Sept 2012 <i>Giza, Egypt</i>

Teaching Experience

Assistant Professor, ZU/UH/ECU

- Analog and Mixed-Signal IC Design
- Electronic Circuits and Basic Electronics
- Digital Circuit Design
- Electronic Design Automation (EDA)
- RF Microelectronics
- Electrical/Electronic Measurements
- Internet of Things (IoT)

Postdoctoral Fellow (Teaching Contribution), KFUPM

- **Introduction to Research (RES 100)** – course instruction covering research methodology, scholarly literature, research ethics, technical writing, and research communication, 2026.
- **Integrated Circuits Analysis and Design** – contributed to Cadence-based IC design laboratory sessions.

Student Supervision

- Supervised five graduate students (three Master of Science and two Doctor of Philosophy) in analog and mixed-signal IC design and biomedical systems, contributing to research publications.

Education and Training

Ph.D. in Electronic Science and Technology

Sept 2016 – Sept 2021

Shanghai Jiao Tong University (QS: 36), China

Thesis: *On-Chip Implementation of a Power-Efficient Analog Gaussian Kernel Neuron for Artificial Neural Network Classifier*

M.Sc. in Electronics and Communications Engineering

Sept 2011 – Mar 2014

Zagazig University, Egypt (GPA: 3.90/4.00)

Thesis: *Current Conveyor Circuit Based on CMOS Converter for Low Voltage and Low Power Applications*

B.Sc. in Electronics and Communications Engineering

Sept 2004 – May 2009

Zagazig University, Egypt (Excellent with Honors 86.49%, Ranked #1)

Professional Training and Certifications

- **Research Bioethics Certificate of Completion**, National Committee of Bioethics (NCBE), King Abdulaziz City for Science and Technology (KACST), Saudi Arabia, August 2026; completed the NCBE Research Bioethics online course. Valid through August 2029.
- **IEEE T14 – Leveraging IEEE DataPort and IEEE Data Descriptions for Enhanced Research Impact and Student Engagement**, IEEE Educational Activities, May 2026, 4 Professional Development Hours (PDH). [Digital Credential](#).
- **Mastering Edge AI Design Flows Workshop**, Interdisciplinary Research Center for Communication Systems and Sensing (IRC-CSS), KFUPM, 4-day intensive program, 2025.

Journal Publications

* *Corresponding author*

1. **A. R. Mohamed*** and M. A. Al-Absi, "SILO: A Spectre-Driven Simulator-in-the-Loop Framework for Variability-Aware RFIC Optimization and Verification" *IEEE Transactions on Computer-Aided Design of Integrated Circuits & Systems (TCAD)*, 2026. **(Submitted)**
2. **A. R. Mohamed***, A. Al-Khulaifi, and M. A. Al-Absi, "A Generic M-Phase, N-Stage Folded Cross-Coupled CMOS Rectifier Architecture for Low-Power RF Energy Harvesting," *Results in Engineering*, vol. 30, Art. no. 110283, 2026.
3. **A. R. Mohamed***, A. A. Al-Khulaifi, and M. A. Al-Absi, "A High-Efficiency Three-Phase CMOS RF–DC Rectifier for Low-Power IoT Applications," *Computers & Electrical Engineering*, vol. 131, Art. no. 110946, 2026.
4. **A. R. Mohamed** and M. A. Al-Absi*, "Readout Architectures for Resistive Sensors: Metrics, Comparison, and Future Directions," *Arabian Journal for Science and Engineering*, 2026.
5. M. S. Elhadidy, A. T. Elgohr, M. A.A. Mousa, **A. R. Mohamed**, and H. A. Dahab, "Path Planning for 6 DOF Robotic Arm Based on Novel Whale Genetic Algorithm Compared to Fifth Polynomial Planning," *Array*, vol. 30, Art. no. 100916, 2026.
6. **A. R. Mohamed*** and M. A. Al-Absi, "Co-Design of Silicon-Based CMOS Rectifiers and Impedance Matching Networks for Low-Power IoT Systems: Trends and Insights," *IEEE Access*, vol. 13, pp. 183855–183885, 2025.
7. **A. R. Mohamed*** and M. A. Al-Absi, "Hardware Implementation of a Configurable and Power-Efficient Current Conveyor Architecture," *Results in Engineering*, vol. 27, Art. no. 106105, 2025.
8. **A. R. Mohamed*** and M. A. Al-Absi, "Realization of a Resistor-Less CMOS Super Capacitor-Multiplier Using Modified Current Conveyors," *e-Prime – Advances in Electrical Engineering, Electronics and Energy*, vol. 13, Art. no. 101044, 2025.
9. **A. R. Mohamed***, A. Al-Khulaifi, and M. A. Al-Absi, "An Efficient CMOS RF-DC Rectifier Based on a Dual Dynamic Self-Biasing Approach," *IEEE Access*, vol. 13, pp. 86841–86847, 2025.
10. **A. R. Mohamed*** and M. A. Al-Absi, "A Large Active Inductor Emulator for Biomedical Applications," *Arabian Journal for Science and Engineering*, 2025.
11. A. G. Tantawy, A. Wahba, H. F. A. Hamed, and **A. R. Mohamed***, "Temperature-Less-Sensitive Transimpedance Amplifier with Background Light Cancellation Loop," *AEU – International Journal of Electronics and Communications*, vol. 190, pp. 1–15, 2025.
12. A. T. Elgohr, M. A. A. Mousa, **A. R. Mohamed**, H. A. Khater, A. Ma'arif, and I. Suwarno, "The Intelligence Behind Robotic Arms: A Deep Dive into Control Evolution," *Journal of Robotics and Control*, vol. 6, no. 3, pp. 1478–1501, Jun. 2025.
13. M. Talaat, **A. R. Mohamed**, B. E. Sedhom, M. A. A. Mousa, D. Song, and A. Alblawi, "A Novel Technology by Using Three-Stage Op-Amp Design Based on Feedback Capacitor for Islanded Mode Microgrid Stability Control," *Journal of Energy Storage*, vol. 99, pp. 1–34, 2024.
14. M. Talaat, **A. R. Mohamed**, B. E. Sedhom, A. Wahba, M. H. Elkholy, N. M. Elbehairy, D. Song, and T. Senjyu, "Monolithic Design of Self-Adaptive CMOS Converter and Robust Event-Triggered Consensus Control for Integration of Multi-Renewable Energy Sources with Battery Storage System," *Journal of Energy Storage*, vol. 88, pp. 1–21, 2024.
15. A. M. Salem, A. Wahba, H. F. A. Hamed, and **A. R. Mohamed***, "A Pseudo Resistor with Temperature Self-Adaptive Scheme," *Integration, the VLSI Journal*, pp. 1–12, 2024.

16. A. Hosny, F. A. Farag, A. Wahba, and **A. R. Mohamed***, "A Power-Efficient Dynamic-Time Current Mode Comparator," *AEU – International Journal of Electronics and Communications*, vol. 171, pp. 1–12, 2023.
17. **A. R. Mohamed***, L. Qi, and G. Wang, "A Power-Efficient and Re-Configurable Analog Artificial Neural Network Classifier," *Microelectronics Journal*, vol. 111, pp. 1–10, 2021.
18. **A. R. Mohamed***, L. Qi, Y. Li, and G. Wang, "A Generic Nano-Watt Power Fully Tunable 1-D Gaussian Kernel Circuit for Artificial Neural Network," *IEEE Trans. Circuits Syst. II: Express Briefs*, vol. 67, pp. 1529–1533, 2020.
19. **A. R. Mohamed*** and G. Wang, "Temperature Effect and Compensation Scheme for Tunable Ultra-High-Resistance Pseudo Resistor," *Microelectronics Journal*, vol. 102, pp. 1–4, 2020.
20. M. F. Ibrahim, **A. R. Mohamed**, F. Farag, and I. L. Abdalla, "An Automated Design Algorithm–Simulated Annealing-Based for Design Analog Circuits," *International Journal of Advanced Science and Technology*, vol. 29, pp. 14023–14035, 2020.
21. **A. R. Mohamed***, M. Farahat, and F. Farag, "Input-Output Rail-to-Rail CMOS CCII for Low Voltage and Low Power Applications," *Microelectronics Journal*, vol. 48, pp. 60–75, 2016.

Conference Publications

1. **A. R. Mohamed***, "Power-Efficient Gm–C Band-Pass Filter for Bio-Signal Acquisition," in *Proc. 2026 IEEE Int. Symp. Circuits Syst. (ISCAS)*, Shanghai, China, 2026, pp. 60–64.
2. M. A. Al-Absi and **A. R. Mohamed***, "Design and Characterization of a Configurable Current Conveyor Circuit," in *Proc. Int. Conf. Microelectronics (ICM)*, Doha, Qatar, 2024, pp. 1–4.
3. F. A. Farag, **A. R. Mohamed**, and A. Wahba, "A CMOS Two-Stage Amplifier Design Methodology for CAD Tools," in *Proc. Int. Conf. Electronic Engineering (ICEEM)*, Egypt, 2021, pp. 1–4.
4. **A. R. Mohamed***, M. Chen, and G. Wang, "Untrimmed CMOS Nano-Ampere Current Reference with Curvature-Compensation Scheme," in *Proc. IEEE Int. Symp. Circuits Syst. (ISCAS)*, Japan, 2019, pp. 1–4.
5. A. Wahba, **A. R. Mohamed**, Y. Khalaf, and F. Farag, "Differential Input–Differential Output CMOS Current Conveyor for High Frequency Applications," in *Proc. WCCAIS*, Tunisia, 2015, pp. 1–6.
6. **A. R. Mohamed***, M. Farahat, and F. Farag, "Input Offset Cancellation Trimming Technique for Operational Amplifiers," in *Proc. SIEPCPC*, Fira, 2013, pp. 1–5.
7. **A. R. Mohamed***, M. Farahat, and F. Farag, "Automated Design Technique for Constant- g_m Rail-to-Rail Input Stage," in *Proc. Int. Conf. Microelectronics (ICM)*, Algiers, 2012, pp. 1–4.
8. **A. R. Mohamed*** and A. Wahba, "CMOS Two-Stage Amplifier Design Approach," in *Proc. WSEAS Conf. Latest Trends on Circuits*, Greece, 2010, pp. 21–24.

Patents

1. M. A. Al-Absi and **A. R. Mohamed**, "A New Configurable Current Conveyor Circuit," KFUPM Ref. #2024-316.

2. M. A. Al-Absi and **A. R. Mohamed**, “Significantly Huge Positive and Negative Capacitor Multiplier,” KFUPM Ref. #2024-318.
3. M. A. Al-Absi and **A. R. Mohamed**, “Realization of a Large Active Inductor for Biomedical Applications,” KFUPM Ref. #2025-105.
4. M. A. Al-Absi and **A. R. Mohamed**, “A New Wide Dynamic Range and Efficient CMOS RF-to-DC Converter Using Dual Biasing Scheme,” KFUPM Ref. #2025-231.
5. M. A. Al-Absi and **A. R. Mohamed**, “Three-Phase and High Efficiency CMOS RF-DC Converter for IoTs and Wireless Nodes Applications,” KFUPM Ref. #2025-290.
6. M. A. Al-Absi and **A. R. Mohamed**, “A Multi-Phase, Multi-Stage Folded Cross-Coupled CMOS Rectifier for RF Energy Harvesting Applications,” KFUPM Ref. #2025-620.
7. **A. R. Mohamed** and M. A. Al-Absi, “Spectre-Based Loop Optimization Workflow for Robust Analog Circuit Sizing with Automated Verification and Validation,” KFUPM Ref. #2026-186.

Honors, Awards, and Grants

1. IEEE ISSCC Student Research Preview (SRP) Poster Award, 2022.
2. KFUPM Postdoctoral Excellence Award (Runner-up, Distinction), 2025–2026.
3. Certificate of Appreciation for serving as a Volunteer in the 2025 IEEE SSCS Arduino Contest, IEEE Solid-State Circuits Society (SSCS), 2025.
4. Certificate of Appreciation for serving as a session chair at ICEEE and supporting the organization of conference sessions, 2025.
5. Certificate of Appreciation for contributions to the Mawhiba Academic Enrichment Program 2025 at KFUPM (Electronics Laboratory assistant; training and supervision support).
6. Certificate of Presentation for author contributions to papers presented at ICM 2024 (Doha) and ICEEE 2025 (Istanbul).
7. Grant from XFAB and BICASL to fabricate integrated chips (2016–2020).
8. Grant from Shanghai Jiao Tong University to pursue the Ph.D. degree (2016–2021).
9. Grant from MOSIS Company under MOSIS Education Program (MEP) to fabricate a chip (2013–2014).
10. Ideal Teaching Assistant, Electronics and Communication Engineering Department (2010–2014).
11. ITIDA grant for graduation project through collaboration between companies, universities and research centres (2009).
12. Ranked #1 at ECE Department and 4th at the college level (2009).

Presentations

1. Research Project Presenter, “Power-Efficient Analog Interface Circuit for Multi-Sensor IoT Nodes in Cognitive Cities,” IRC-CSS Research Showcase Day, King Fahd University of Petroleum & Minerals (KFUPM), Dhahran, Saudi Arabia, August 16, 2026.

2. Invited Seminar Speaker, "From Optimization to Signoff: An Automated Spectre-Integrated Workflow for Robust CMOS Analog Circuit Design with Verification and Validation," Electrical Engineering Department, IRC for Communication Systems and Sensing, KFUPM, Saudi Arabia, March 31, 2026.
3. Invited Seminar Speaker, "Silicon-Based CMOS Integrated Circuits for Low-Power Sensing Applications," Department of Electrical Engineering, College of Engineering and Physics, KFUPM, Dhahran, Saudi Arabia, Nov. 5, 2025.
4. Presenter, IEEE International Conference on Electrical and Electronics Engineering (ICEEE 2025), Istanbul.
5. Presenter, "Design and Characterization of a Configurable Current Conveyor Circuit," ICM 2024, Doha, Qatar.
6. Presenter, "A Power-Efficient and Re-Configurable Mixed Mode Artificial Neural Network Classifier," ISSCC-SRP 2021, USA.
7. Presenter, "A Generic Nano-Watt Power Fully Tunable 1-D Gaussian Kernel Circuit," ISI-CAS 2020, China.
8. Presenter, "Untrimmed CMOS Nano-Ampere Current Reference with Curvature-Compensation Scheme," ISCAS 2019, Japan.
9. Lecturer, "Solar Energy System," High Institute of Technology, Zagazig, Egypt, 2015.
10. Lecturer, "Arduino Workshop," Zagazig University (ASC activities), 2015.
11. Lecturer, "FPGA and VHDL," IEEE Robotics and Automation Society (RAS), Zagazig Student Branch, 2014.
12. Talk, "Mini-Electronic Projects," IEEE-ZSB, Zagazig University, 2014.
13. Thesis Presentation, Zagazig University, 2014.
14. Presentation, "IC Layout Design," Zagazig University, 2012 and 2013.
15. Presentation, Association of Egyptian American Scholars 37th Annual Int. Conf., Zagazig University, 2010.
16. Presentation, 4th forum of ITAC research cooperation initiative, Egypt, 2010.
17. Presentation, Egyptian Engineer Day (EED), graduation project, Nasr City, Egypt, 2009.

Selected Projects

1. **Project No. IN26115: Power-Efficient Analog Interface Circuit for Multi-Sensor IoT Nodes in Cognitive Cities (2026–2029)**
NexusAFE – Configurable Ultra-Low-Power Multi-Sensor Analog Interface Platform. Research Team Member / Co-PI, KFUPM Deanship of Research-funded project (total project funding: SAR 356,500). Development of a configurable CMOS/COTS analog front-end platform integrating PVT-robust analog circuits, time-based conversion, wireless sensing, synchronized acquisition, and application-level validation.
2. **Project No. INCS2606: Prototype of a Low-Power Wearable IoT-Based Multichannel Hemodynamic Monitoring System (2026–2027)**
HEMOX – Multichannel Hemodynamic Monitoring Platform. Research Team Member / Co-PI, IRC-CSS-funded project (total project funding: SAR 50,000). Development of a synchronized dual-site, dual-wavelength PPG sensing platform with low-power wireless acquisition, motion- and signal-quality-aware processing, physiological-parameter estimation, and real-time analysis through HEMOX Explorer.

3. **Project No. INCS2501: Design and Simulation of an Integrated RF–DC Rectifier for Energy Harvesting** (2025–2026)
Research Team Member / Co-PI, IRC-CSS-funded project (total project funding: SAR 63,300). Development of high-efficiency CMOS RF–DC rectifier architectures for low-power IoT and energy-harvesting applications, including self-biasing, multiphase, and multistage rectifier topologies.
4. **SILO – Spectre-Driven Simulator-in-the-Loop Framework for Variability-Aware Analog/RFIC Optimization and Verification**
Development of an automated Cadence Spectre-integrated design framework combining circuit optimization, PVT-corner analysis, Monte Carlo variability assessment, design verification, and robustness evaluation.
5. **Hardware Community Leader: IoT-Based Healthcare System Development** (2025–Present)
Leading hardware-oriented research and prototype development for IoT-based sensing, wearable systems, and healthcare applications.
6. **On-Chip Analog/Mixed-Signal Artificial Neural Network Implementation** (2017–2021)
Design and implementation of power-efficient CMOS analog neural-network building blocks, including tunable Gaussian-kernel neurons and mixed-signal classifier architectures.
7. **Analog Front-End for PPG Acquisition System** (Supervisor, 2023–2024)
Supervision of analog front-end circuit design for photoplethysmography signal acquisition and biomedical sensing applications.
8. **Power Management Integrated Circuit** (Supervisor, 2022–2023)
Supervision of CMOS power-management circuit design for low-power electronic systems.

Additional Projects:

- On-chip implementation of analog neuron circuits, low-dropout voltage regulators, successive-approximation-register analog-to-digital converters, bandgap reference circuits, amplifiers, filters, voice activity detectors, level shifters, and CMOS current sources.
- PRBS generators for high-speed transceivers, robotic systems, and RFID-based control systems.
- FPGA-based MIPS processor, steganography algorithms using MATLAB, and ASIC switching regulators.

Personal Skills

Mother Tongue: Arabic

English: IELTS 6.0; (British Council certificate)

Professional Memberships and Service

- Member, Institute of Electrical and Electronics Engineers (IEEE)
- Volunteer Judge, IEEE Solid-State Circuits Society (SSCS) Arduino Contest, 2025 and 2026
- Session Chair, IEEE International Conference on Electrical and Electronics Engineering (ICEEE), 2025
- Member, Engineering Syndicate of Egypt
- Member, Activities of Science Clubs
- Member, Zagazig Student Branch activities (student skills development)
- Volunteer, Resala Charity

Computer Skills and Competences

Technology: IC design, fabrication, and testing using IBM 0.13 μ m, XFAB 0.18/0.35 μ m, and SKYWater/SKY130.

Software: Cadence flow, Virtuoso Layout, Skill scripting, Calibre, OrCAD, Proteus, ISE, Visual Studio, L-Edit, Circuit Maker, CodeVision, MATLAB, C/C++, Python, Verilog, VHDL, LabVIEW, Altium, Xschem, Ngspice, Magic, open-source tools.

Hardware: FPGA (Spartan3E), Atmel microcontrollers (C), PCB, Arduino/ESP.

Microsoft Office: Word, Excel, PowerPoint, Outlook.

Academic Referees

Prof. Munir A. Al-Absi — Professor, Department of Electrical Engineering, King Fahd University of Petroleum and Minerals (KFUPM), Dhahran 31261, Saudi Arabia. Email: mku-laib@kfupm.edu.sa, Phone: +966-3-860-3696

Prof. Wang Guoxing — Professor, Department of Micro-Nano Electronics, Shanghai Jiao Tong University, China; Secretary of the Party Branch, Department of Micro-Nanoelectronics; Vice Chairman, IEEE Circuits and Systems Society; Chief Editor, IEEE Transactions on Biomedical Circuits and Systems. Email: guoxing@sjtu.edu.cn

Dr. Yongfu Li — Associate Professor, Department of Micro-Nano Electronics, Shanghai Jiao Tong University, China; IEEE Circuits and Systems Society Board of Governors (Region 10); Associate Editor, IEEE Transactions on Biomedical Circuits and Systems. Email: yongfu.li@sjtu.edu.cn

Dr. Qi Liang — Assistant Professor, Department of Micro-Nano Electronics, Shanghai Jiao Tong University, China. Email: qi.liang@sjtu.edu.cn

Dr. Mahmoud Ali — Assistant Professor, School of Mathematical and Computer Sciences, Heriot-Watt University Dubai, UAE. Email: M.Mousa@hw.ac.uk, Phone: +971-502480694

Dr. Shaimaa Ahmed Elsaid — Associate Professor, Department of Computer Engineering, College of Computer Engineering and Sciences, Prince Sattam bin Abdulaziz University, Al-Kharj 16278, Saudi Arabia. Email: sh.ahmed@psau.edu.sa, Phone: +966-570147566